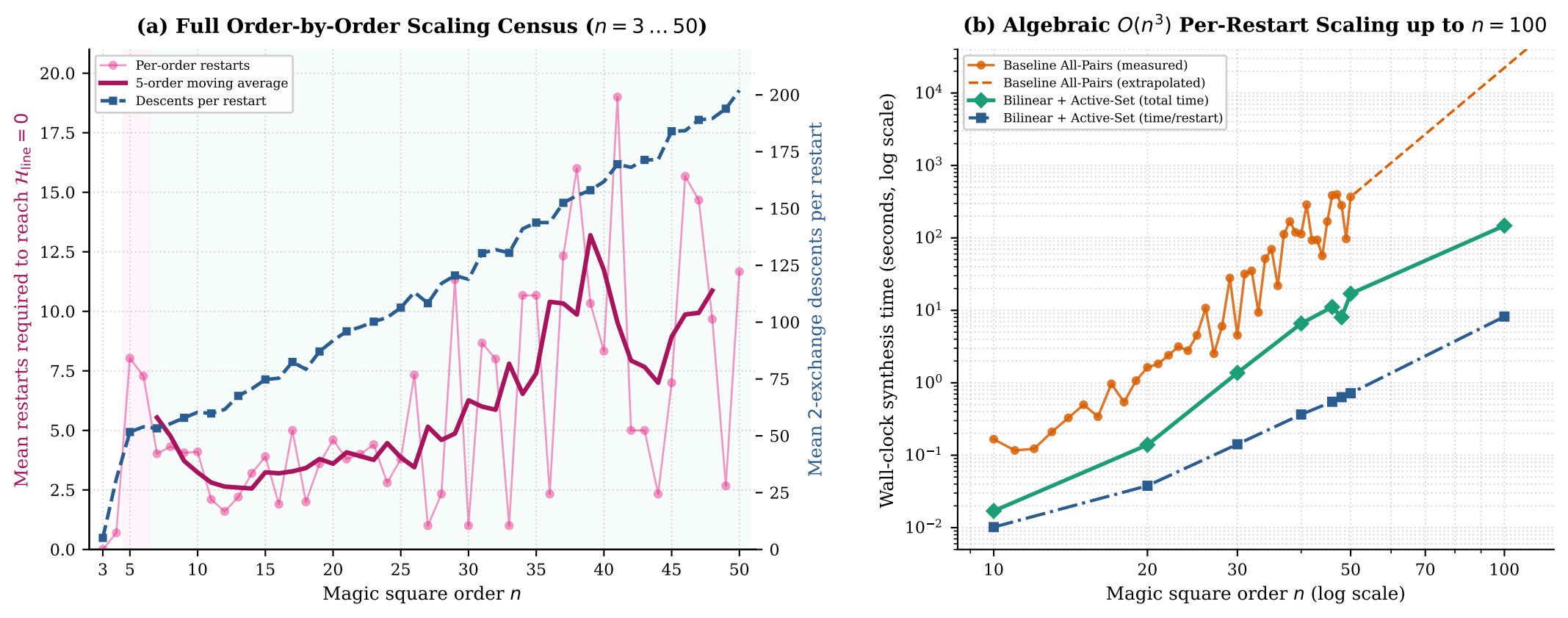
(a) Full Order-by-Order Scaling Census ($n = 3 \ldots 50$)
Per-order restarts
5-order moving average
Descents per restart
Mean restarts required to reach $\mathcal{H}_{\text{line}} = 0$
Mean 2-exchange descents per restart
Magic square order n
(b) Algebraic $O(n^3)$ Per-Restart Scaling up to $n = 100$
Baseline All-Pairs (measured)
Baseline All-Pairs (extrapolated)
Bilinear + Active-Set (total time)
Bilinear + Active-Set (time/restart)
Wall-clock synthesis time (seconds, log scale)
Magic square order n (log scale)